Sequences and Arrays with Perfect Periodic Correlation

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Properties and methods for synthesizing sequences with perfect periodic autocorrelation functions and good energy efficiency are discussed. The construction is extended to two-dimensional perfect arrays. The construction methods used are based mainly on a search in the frequency domain and on a multiplication theorem for periodic sequences and arrays.

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I. INTRODUCTION

Time-discrete signals with good autocorrelation properties are useful for many applications in communication and measurement. On principle it is not possible to find sequences with an ideal impulse-like aperiodic autocorrelation function. On the other hand, the periodic repeated autocorrelation function (PACF) can disappear between its periodic main peaks. Sequences of this property are called perfect sequences [1–3]. For most applications, sequences with good aperiodic autocorrelation properties are of interest, but no methods exist to construct optimal binary or ternary sequences of this kind. On the other hand, it is relatively simple to find perfect sequences with an ideal PACF. There are two connections between both problems. Sometimes sequences with ideal PACF also show good aperiodic autocorrelation properties [4]. Another way is to transmit a perfect sequence twice and then to correlate it with the single sequence. The resulting correlation function then ideally disappears between the two main peaks. In this window region, for example, the impulse response of a transmitting channel may be measured and used for adjusting an adaptive equalizer, whereas the main peaks are useful for synchronization purposes. Multiple sequences of this kind are useful for radar applications [5]. Perfect sequences are also used for the synthesis of orthogonal matrices for source coding or in complementary coding, [5, chap. IV].

The concept of perfect sequences can readily be extended to higher dimensions. Two-dimensional perfect arrays are fundamental in coded aperture imaging, e.g., phased array antennas, arrays of sound sources or flashing tomography [6–11]. Further, arrays are used in higher-dimensional signal processing applications such as time-frequency-coding [12], spatial correlation or map-matching [13–15], parallel processing in one-dimensional correlation [16], arrays for built-in tests of VLSI-circuits [17] or two-dimensional measuring techniques, e.g., optical systems.

II. PERFECT SEQUENCES WITH GOOD ENERGY EFFICIENCY

A. Correlation Properties Of Sequences

Let $s_i(n)$ be real sequences with length N. Their periodic repetition with period N gives the periodic sequences $\tilde{s}_i(n)$. The periodic autocorrelation functions $\tilde{\varphi}_{ii}(m)$ and, respectively, the crosscorrelation functions $\tilde{\varphi}_{ij}(m)$ are given by

$$\underset{j}{\tilde{\varphi}_{ii}}(m) = \sum_{n=0}^{N-1} \tilde{s}_i(n)\, \underset{j}{\tilde{s}}(n+m). \tag{1}$$

The sequence is called perfect if one period of the PACF gives

$$\tilde{\varphi}_{ii}(m) = \begin{cases} E, & \text{for } m = 0 \\ 0, & \text{for } m \neq 0 \end{cases},$$

$$\text{in the region } 0 \leq m < N \quad (2)$$

with the energy E of the sequence given by

$$E = \sum_{n=0}^{N-1} s_i^2(n). \quad (3)$$

The discrete Fourier transform (DFT) of $\tilde{s}_i(n)$ generates a discrete periodic spectrum of period N

$$\tilde{S}_i(k) = \sum_{n=0}^{N-1} \tilde{s}_i(n) \exp(-j2\pi nk/N), \qquad 0 \leq k < N. \quad (4)$$

Then by combining (1) and (4) the DFT for the PACF is given by

$$\tilde{\varphi}_{ii}(m) \leftrightarrow |\tilde{S}_i(k)|^2 = E \quad (5)$$

therefore the magnitude of the spectrum of perfect sequences is always the constant $\sqrt{E}$ [18, 19]. Further a perfect sequence cannot be DC free, as setting k to zero in (4) and (5) gives

$$|\tilde{S}_i(0)| = \left| \sum_{i=0}^{N-1} s_i(n) \right| = \sqrt{E}. \quad (6)$$

B. Product Theorem For Periodic Correlation Functions

Consider a set of periodic sequences $\tilde{s}_i(n)$, $\tilde{s}_j(n)$, ..., of period N_1 with the periodic autocorrelation and crosscorrelation functions $\tilde{\varphi}_{ii}(m)$, $\tilde{\varphi}_{ij}(m)$. Also consider the periodic sequences $\tilde{g}_i(n)$, $\tilde{g}_j(n)$ of period N_2 with the correlation functions $\tilde{\zeta}_{ii}(m)$, $\tilde{\zeta}_{ij}(m)$. If the periods N_1 and N_2 are relatively prime (that means that their greatest common divisor is 1), then the product sequences

$$\tilde{p}_{\substack{i \\ j}}(n) = \tilde{s}_{\substack{i \\ j}}(n)\, \tilde{g}_{\substack{i \\ j}}(n) \quad (7)$$

are periodic with least period $N_1 \cdot N_2$.

The correlation functions of these product sequences $\tilde{\sigma}_{ii}(m)$, $\tilde{\sigma}_{ij}(m)$ can be expressed as

$$\tilde{\sigma}_{ij}(m) = \sum_{n=0}^{N_1 N_2 - 1} \tilde{p}_i(n)\, \tilde{p}_j(n+m) \quad (8)$$

and using (7)

$$= \sum_{n=0}^{N_1 N_2 - 1} \tilde{s}_i(n)\, \tilde{g}_i(n)\, \tilde{s}_j(n+m)\, \tilde{g}_j(n+m).$$

Since N_1 and N_2 are relatively prime, in this expression all possible products $\tilde{s}_i(n)\, \tilde{s}_j(n+m)$ or $\tilde{g}_i(n)\, \tilde{g}_j(n+m)$ appear only once in a period $N_1 N_2$. Therefore this sum of $N_1 N_2$ quadruple products can be written as

$$\tilde{\sigma}_{ij}(m) = \left[\sum_{n=0}^{N_1 - 1} \tilde{s}_i(n)\, \tilde{s}_j(n+m) \right] \times \left[\sum_{n=0}^{N_2 - 1} \tilde{g}_i(n)\, \tilde{g}_j(n+m) \right]$$

$$\tilde{\sigma}_{ij}(m) = \tilde{\varphi}_{ij}(m)\, \tilde{\zeta}_{ij}(m). \quad (9)$$

That is, the periodic autocorrelation and crosscorrelation functions of the product of periodic sequences of relatively prime lengths are themselves the products of the individual correlation functions [20].

In particular for the case $i = j$ the autocorrelation function of the product of two sequences with relatively prime periods can be expressed as the product of both periodic autocorrelation functions [21]. Examples are given below.

The use of (9) for the synthesis of sets of sequences with vanishing periodic crosscorrelation functions is shown in [20].

C. Synthesis Of Perfect Sequences

Each perfect sequence $\tilde{s}_i(n)$ possesses a DFT $\tilde{S}_i(k)$ of constant discrete magnitude, (5). Therefore combining a constant magnitude spectrum with any odd-symmetrical phase spectrum

$$\tilde{\psi}_i(N-k) = \tilde{\psi}_i(k), \qquad \text{in the region } 0 \leq k < N \quad (10)$$

always gives by inverse DFT a real, perfect sequence [19]. For most applications the sequences should possess a good energy efficiency η [22] (or equivalently, a low peak factor $1/\eta$), defined by

$$\eta = \sum_{n=0}^{N-1} s^2(n) / (N \max[s^2(n)]). \quad (11)$$

Probably only one sequence with the maximum possible energy efficiency $\eta = 1$ exists [23]:

$$s(n) = \{1, \quad 1, \quad 1, \quad -1\}. \quad (12)$$

No method for synthesizing real, perfect sequences with best energy efficiencies is currently known. The results of a computer search are shown in Table I and Fig. 1. At first this search was conducted up to length $N = 24$. The results showed that sequences with best energy efficiency and even period almost all possessed a phase spectrum which was even-symmetrical in the region $0 < k < N/2$. Using this symmetry conjecture the search could be carried on up to $N = 30$ with good results (Table I). As a second method, the product theorem (9) was used to combine two perfect sequences of relatively prime lengths N_1, N_2 into a new sequence of length $N_1 N_2$. This product sequence always is perfect, because the product of ideal PACFs of relatively prime periods is ideal.

The energy efficiency of the product sequence can be shown to be the product of the single energy efficiencies [20]

$$\eta_{12} = \eta_1 \eta_2. \quad (13)$$

TABLE I
Perfect Sequences With Best Energy Efficiencies

N	type	η	Sequences of good energy efficiency ($\eta \geq 60\%$)
3	S,B	75	1 1 a (a = -0.50)
4	B	100	1 1 1 $\bar{1}$
5	S	67	1 $\bar{1}$ .33 1 .5
6	S,T	67	1 0 $\bar{1}$ 1 0 1
7	B	72	1 1 1 a 1 a a (a = -0.586)
8	S	69	1 1 .17 1 $\overline{.17}$ 1 $\bar{1}$ $\overline{.66}$
9	S	71	1 1 $\overline{.45}$ 1 1 .68 $\overline{.85}$ .12 $\overline{.98}$
10	S	66	1 $\bar{1}$ $\overline{.87}$ $\overline{.24}$ $\overline{.87}$ 0 $\overline{.87}$ .24 $\overline{.87}$ 1
11	S	77	1 $\bar{1}$ 1 .90 $\overline{.12}$ $\bar{1}$ .46 $\bar{1}$ .70 1 1
"	B	73	1 1 a 1 1 1 a a a 1 a (a = -0.634)
12	P	75	4 ⊙ 3
13	S	79	1 .94 1 $\bar{1}$ $\bar{1}$ .47 $\bar{1}$ 1 .47 .47 .94 .93 $\bar{1}$
"	T	69	1 0 1 0 0 1 $\bar{1}$ 1 0 $\bar{1}$ $\bar{1}$ 1 1
14	S	75	1 $\overline{.38}$ $\overline{.84}$ $\bar{1}$ 1 0 1 1 $\overline{.84}$ .38 1 1 .93 $\bar{1}$
15	B	74	1 1 1 1 a 1 a 1 1 a a 1 a a a (a = -0.667)
"	S	80	1 1 1 1 1 $\overline{.8}$ .15 1 $\overline{.92}$ .69 $\bar{1}$ .95 .38 $\overline{.98}$ $\bar{1}$
16	S	80	1 1 $\overline{.1}$ 1 .8 $\bar{1}$ $\overline{.1}$ $\bar{1}$ 1 $\overline{.9}$ 1 .9 $\bar{1}$ $\overline{.9}$ 1 .9
17	S	72	1 1 $\bar{1}$ $\overline{.98}$ $\overline{.12}$ $\overline{.66}$ .55 .61 .93 $\overline{.37}$ 1 $\bar{1}$ .68 1 1 .86 $\bar{1}$
"	L	65	1 1 a 1 a a a 1 1 a a a 1 a 1 1 .2 (a = -0.61)
18	S	73	1 $\overline{.1}$ .42 .53 .86 1 $\overline{.96}$ 1 1 0 1 $\bar{1}$ $\overline{.96}$ $\bar{1}$ .86 $\overline{.53}$ .42 1
19	B	75	1 1 a a 1 1 1 1 a 1 a 1 a a a a 1 1 a (a = -0.691)
"	S	82	1 1 1 $\bar{1}$ 1 .45 1 $\overline{.93}$ $\overline{.93}$ $\bar{1}$ $\overline{.63}$ 1 .45 .45 1 $\overline{.93}$ 1 1 $\bar{1}$
20	S	74	1 $\overline{.98}$ $\overline{.96}$ 0 $\overline{.07}$ $\bar{1}$ 1 .76 .95 1 1 $\bar{1}$ .95 $\overline{.76}$ 1 1 $\overline{.07}$ .02 $\overline{.96}$ .98
21	T	76	1 1 1 1 1 $\bar{1}$ 1 0 1 0 $\bar{1}$ 1 1 $\bar{1}$ 0 0 1 $\bar{1}$ 0 $\bar{1}$ $\bar{1}$
22	S	72	1 $\bar{1}$ $\overline{.94}$ $\overline{.68}$ .92 $\overline{.85}$ 1 1 .05 0 .05 $\bar{1}$ 1 .85 .92 .68 $\overline{.94}$ 1 1 1 $\overline{.08}$ $\bar{1}$
23	B	76	1 1 1 1 1 a 1 a 1 1 a a 1 1 a a 1 a 1 a a a a (a = -0.71)
24	S	67	1 $\overline{.67}$ .33 1 $\bar{1}$ $\overline{.67}$ $\bar{1}$ $\overline{.82}$ 1 .33 .33 $\overline{.82}$ 1 $\overline{.67}$ 1 1 1 $\overline{.67}$ .33 .82 $\bar{1}$ .33 1 .82
26	S	68	1 $\overline{.47}$ $\overline{.44}$ $\overline{.95}$ 1 1 $\overline{.23}$ $\overline{.95}$ $\overline{.70}$ $\bar{1}$ 1 $\overline{.81}$ .98 0 .98 .81 1 1 $\overline{.70}$ .95 $\overline{.23}$ $\bar{1}$ 1 .95 $\overline{.44}$ .47
28	P	72	4 ⊙ 7
29	L	71	1 a a 1 1 1 1 a 1 a a a 1 a a 1 a a a 1 a 1 1 1 1 a a 1 .16 (a = -0.687)
30	S	74	1 $\bar{1}$ $\overline{.44}$ $\bar{1}$ 1 1 $\overline{.81}$ $\overline{.77}$ $\overline{.78}$ .06 1 $\overline{.97}$ 1 $\overline{.94}$ .90 0 .90 .94 1 .97 1 $\overline{.06}$ $\overline{.78}$ .77 $\overline{.81}$ $\bar{1}$ 1 1 $\overline{.44}$ 1
31	T	81	1 0 1 1 1 $\bar{1}$ 0 $\bar{1}$ $\bar{1}$ $\bar{1}$ 0 1 $\bar{1}$ 1 1 $\bar{1}$ 0 1 $\bar{1}$ 1 $\bar{1}$ 1 1 1 1 $\bar{1}$ $\bar{1}$ 1 1 1 0 0
35	B	79	1 a 1 1 1 a a a 1 1 1 1 1 a 1 1 1 a a 1 a a a a 1 a 1 a 1 1 a a 1 a a (a ≈ -0.75)

TABLE I (Continued)

N	type	η	Sequences
36	P	71	4 ⊙ 9
37	L	74	1 a 1 1 a a 1 a 1 1 1 1 a a a 1 a a a a 1 a a a
			1 1 1 1 a 1 a a 1 1 a 1 .14 (a = -0.718)
41	L	75	1 1 a 1 1 a a 1 1 1 a a a a a 1 a 1 a 1 1 a 1 a 1
			a a a a a 1 1 1 a a 1 1 a 1 1 .135 (a ◻ -0.73)
43	B	80	1 1 a a 1 a 1 a a 1 1 1 a 1 1 1 1 1 a a a 1 a 1 1 1 a a a a
			a 1 a a a 1 1 a 1 a 1 1 a (a = -0.768)
44	P	77	4 ⊙ 1 1
47	B	81	1 a a a a 1 a a a a 1 1 a 1 a 1 a a a 1 1 a 1 1 a a
			1 a a 1 1 1 a 1 a 1 a a 1 1 1 1 a 1 1 1 1 (a ◻ - 0.776)
48	P	60	3 ⊙ 1 6
52	P	79	4 ⊙ 1 3
53	L	77	1 a a 1 a 1 1 a 1 1 1 a 1 a 1 1 1 a a a a a a
			1 1 a a 1 1 a a a a a a 1 1 1 a 1 a 1 1 1
			a 1 1 a 1 a a 1 .12 (a ◻ -0.758)
57	T	86	1 0 1 1 1 1 $\bar{1}$ $\bar{1}$ $\bar{1}$ 1 1 0 1 1 1 $\bar{1}$ $\bar{1}$ $\bar{1}$ 1 1 $\bar{1}$ 1 $\bar{1}$ 1 1 1 1 1 $\bar{1}$ 1
			0 $\bar{1}$ 1 $\bar{1}$ 0 $\bar{1}$ $\bar{1}$ $\bar{1}$ 1 $\bar{1}$ $\bar{1}$ 0 1 $\bar{1}$ 1 1 $\bar{1}$ $\bar{1}$ 1 1 0 $\bar{1}$ 1 $\bar{1}$ 1 0 0
59	B	82	1 a 1 a a a 1 a 1 a 1 1 a 1 1 a a a 1 a a a a 1 1 a a a a a
			1 1 1 1 1 a a 1 1 1 1 a 1 1 1 a a 1 a a 1 a 1 a 1 1 1 a 1
			(a = - 0.794)
60	P	74	4 ⊙ 15

Note: *B* is binary sequence, *T* is ternary sequence, *L* is Legendre sequence, *S* is computer search, *P* is product sequence $(N_1 \cdot N_2)$, $\overline{.43} = -0.43$.

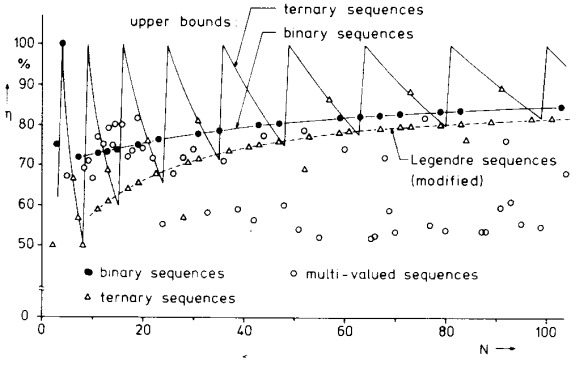


Fig. 1. Energy efficiencies of perfect sequences.

Therefore the energy efficiency of product sequences will mostly become smaller. The only exception to this is the product with the perfect binary sequence of length 4, (12). Product sequences with good energy efficiency are also shown in Table I and Fig. 1.

D. Perfect Binary And Ternary Sequences

In many applications the use of binary or ternary sequences can simplify implementation. Perfect but amplitude-asymmetrical binary sequences can be found by modifying binary pseudonoise (PN) sequences. The PACF of a PN sequence $s(n) \in \{1, -1\}$ has the constant value of -1 between the main peaks [2, 3]. Substituting the values -1 in $s(n)$ by a suitable q, a perfect but amplitude-asymmetrical binary sequence can be constructed [3]. Because each period of any PN sequence can be written with $(N+1)/2$ positive and $(N-1)/2$ negative elements, q can be calculated considering (6) and (2) as

$$q = -1/(1 + 2/\sqrt{N+1}). \tag{14}$$

As $N \to \infty$ then q becomes -1. Therefore very long sequences are nearly amplitude-symmetrical and their energy efficiency tends to 1. The modified perfect PN sequences of lengths $N = 3, 7, 11, 15, 19, 23, 35, \ldots$, are included in Table I and Fig. 1. In the same manner Legendre sequences can be modified. Legendre sequences

are binary sequences but with a trailing zero; they possess a two-valued PACF [6]. By adding a suitable constant, a perfect but amplitude-asymmetrical ternary sequence can be built. Legendre sequences exist for all prime lengths. In Table I, which shows the sequences of best-known energy efficiency, the modified Legendre sequences of lengths $N = 17, 29, 37, 41$, and 53 are contained. The energy efficiency of the modified Legendre sequences is given by $\eta = 1/(1 + 1/\sqrt{N})^2$. It also tends to 1 for $N \to \infty$ as is depicted in Fig. 1.

Perfect ternary sequences with $s(n) \in \{1, 0, -1\}$ have been constructed, e.g., by Chang, Ipatov, Shedd & Sarwate, and Høholdt & Justesen [21, 24–27]. The number of nonzero elements in a perfect ternary sequence is always a square number. This can be seen by considering that following (6), the energy of a perfect sequence equals the square of its mean value. Further, this square number cannot exceed the length N. Therefore the energy efficiency of a perfect ternary sequence is bounded by

$$\eta \leq (\text{entier } \sqrt{N})^2/N \tag{15}$$

(with entier (x) = greatest integer $\leq x$). This sawtooth-like bound is sketched in Fig. 1. It approaches 1 for large N. Ternary sequences reaching this bound can be synthesized using polynomial-algebraic methods in the Galois-field (GF) (3), for the lengths [21, 26]

$$N = (p^{w(2d+1)} - 1)/(p^w - 1) \text{ with } p \text{ prime;}$$
$$d, w \text{ positive integer} \tag{16}$$

(that is $N = 7, 13, 21, 31, 57, 73, 91, \ldots$). The periodic multiplication (9) of two ternary sequences of relatively prime lengths gives more ternary sequences with reduced energy efficiencies. The results of these multiplications are also to be found in Fig. 1 and partly in Table I.

Likewise, the periodic product of two amplitude-asymmetrical binary sequences gives a quaternary sequence, or the product of a ternary sequence with an amplitude-asymmetrical binary sequence results in a quinary perfect sequence (Fig. 1). Finally, it may be remarked that for all lengths there exist complex perfect sequences with energy efficiency 1 [28].

III. PERFECT TWO-DIMENSIONAL ARRAYS

A. Correlation Properties Of Arrays

Let $s_i(x, y)$ be rectangular $N_x N_y$ arrays of real numbers. Their periodic repetition with the periods N_x in x-direction and N_y in y-direction give the periodic arrays $\tilde{s}_i(x, y)$. The correlation properties of such arrays are found by generalizing the correlation properties of periodic sequences in Section II A. The periodic autocorrelation and crosscorrelation functions are defined as in (1)

$$\underset{j}{\tilde{\varphi}_{ii}}(m, n) = \sum_{x=0}^{N_x-1} \sum_{y=0}^{N_y-1} \tilde{s}_i(x, y)\, \underset{j}{\tilde{s}_i}(x + m, y + n). \tag{17}$$

An array is denoted perfect if one period of the PACF is described by

$$\tilde{\varphi}_{ii}(m, n) = \begin{cases} E, & \text{for } (m, n) = (0, 0) \\ 0, & \text{for } (m, n) \neq (0, 0) \end{cases},$$
$$\text{in the region } 0 \leq m < N_x \cap 0 \leq n < N_y \tag{18}$$

whereby E is the energy

$$E = \sum_{x=0}^{N_x-1} \sum_{y=0}^{N_y-1} s_i^2(x, y). \tag{19}$$

The two-dimensional DFT of $\tilde{s}_i(x, y)$ is obtained by

$$S_i(u, v) = \sum_{x=0}^{N_x-1} \sum_{y=0}^{N_y-1} \tilde{s}_i(x, y)$$
$$\times \exp[-j2\pi(xu/N_x + yv/N_y)], \quad u, v \in Z \tag{20}$$

From (18) and (20) it follows that the PACF of a perfect array transforms into

$$\tilde{\varphi}_{ii}(m, n) \leftrightarrow |\tilde{S}_i(u, v)|^2 = E. \tag{21}$$

The magnitude of the DFT of a perfect array is therefore the frequency-independent constant $\sqrt{E}$. Further, analogously to (6), a perfect array cannot be DC free, for

$$|\tilde{S}_i(0, 0)| = \left| \sum_{x=0}^{N_x-1} \sum_{y=0}^{N_y-1} s_i(x, y) \right| = \sqrt{E}. \tag{22}$$

B. Product Theorem For Periodic Arrays

The product theorem (9) for periodic sequences and their correlation functions can be generalized to periodic arrays. Let $\tilde{s}_i(x, y)$, $\tilde{s}_j(x, y)$ be a set of periodic arrays with periods N_x, N_y and correlation functions $\tilde{\varphi}_{ii}(m, n)$, $\tilde{\varphi}_{ij}(m, n)$. Further consider $\tilde{g}_i(x, y)$, $\tilde{g}_j(x, y)$ a second set of arrays with periods M_x, M_y and correlation functions $\tilde{\zeta}_{ii}(m, n)$, $\tilde{\zeta}_{ij}(m, n)$. If now the periods N_x and M_x and also the periods N_y and M_y are relatively prime, then the product arrays

$$\underset{j}{\tilde{p}_i}(x, y) = \underset{j}{\tilde{s}_i}(x, y)\, \underset{j}{\tilde{g}_i}(x, y) \tag{23}$$

are periodic with least periods $N_x M_x$ and $N_y M_y$. In an analogous derivation to Section II B it can be shown that one obtains for the correlation functions $\tilde{\sigma}_{ij}(m, n)$ of the product arrays

$$\tilde{\sigma}_{ij}(m, n) = \tilde{\varphi}_{ij}(m, n)\, \tilde{\zeta}_{ij}(m, n) \tag{24}$$

with periods $N_x \cdot M_x$ and $N_y \cdot M_y$. That is, the periodic autocorrelation and crosscorrelation functions of the product of periodic arrays of relatively prime dimensions are themselves the products of the individual periodic correlation functions. As a simple special case arrays with periods $N_x \cdot N_y$ can be constructed by periodically multiplying two one-dimensional sequences $\tilde{s}_i(x)$ and $\tilde{g}_i(y)$ with periods N_x and M_y, respectively

$$\tilde{p}_{ij}(x, y) = \tilde{s}_i(x) \cdot \tilde{g}_j(y) \tag{25}$$

with period $N_x \cdot M_y$. Setting $N_y = M_x = 1$, and using any N_x, M_y, (24) yields the autocorrelation and crosscorrelation functions of these product arrays

$$\tilde{\sigma}_{ij}(m, n) = \tilde{\varphi}_{ij}(m) \cdot \tilde{\zeta}_{ij}(n) \tag{26}$$

with periods N_x, M_y.

C. Synthesis Of Perfect Arrays

As in Section II C for one-dimensional sequences the DFT of a real perfect array $\tilde{s}_i(x, y)$ can be found by combining the constant discrete magnitude spectrum (21) with any two-dimensional odd-symmetrical phase spectrum

$$\tilde{\psi}_i(u, v) = \tilde{\psi}_i(N_x - u, N_y - v) \tag{27}$$

in the region $0 \leq u < N_x \cap 0 \leq v < N_y$. As a further condition a good energy efficiency

$$\eta = \frac{E}{N_x N_y \max[s_i^2(x, y)]} \tag{28}$$

will be required. Some results of a computer search for small arrays with maximum energy efficiencies are shown in Fig. 2(a). Larger arrays can be obtained by two-dimensional folding of sequences [15] or by periodical multiplication of perfect arrays and sequences. The method of folding a perfect sequence of length N into a perfect array of size $N_x N_y$ (with $N = N_x N_y$ and N_x, N_y relatively prime) is illustrated in Fig. 3. The perfect ternary sequence of length $N = 21 = 3 \cdot 7$ is folded into a perfect array of size $3 \cdot 7$. That is because the periodic repeated sequence is to be found on the diagonals of the periodically repeated array. The energy efficiencies of

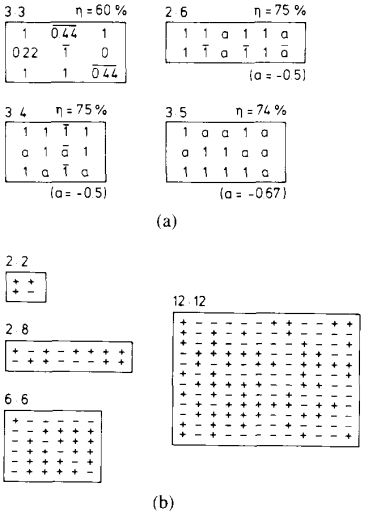


Fig. 2. Perfect arrays of best energy efficiency. a) Multivalued arrays. b) Binary (± 1) arrays.

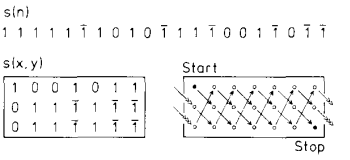


Fig. 3. Folding perfect sequence into perfect array.

folded sequences therefore remain unaltered. The sizes and the energy efficiencies of the folded sequences of Table I are contained in Fig. 4.

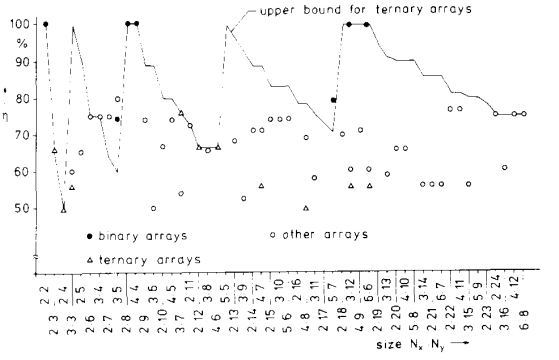


Fig. 4. Energy efficiencies of perfect arrays.

Further, perfect arrays can be constructed by periodical multiplication of perfect sequences and arrays. As an example the periodic product of the arrays $2 \cdot 2$ and $3 \cdot 3$ of Fig. 2 is sketched in Fig. 5. The product array $6 \cdot 6$ is perfect too because 2 and 3 are relatively prime, and using (24) the periodic product of perfect PACFs is itself perfect.

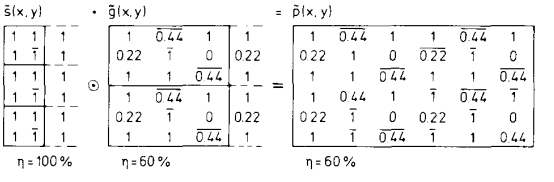


Fig. 5. Multiplication of two perfect arrays of relatively prime dimensions.

For product sequences and also for product arrays the energy efficiencies are multiplicative. Therefore the energy efficiency of product arrays will mostly be smaller. Exceptions are the products with a perfect ± 1 – sequence or array (see Fig 2(b)). The results of this method are included in Fig. 4.

D. Perfect Binary And Ternary Arrays

Perfect binary or ternary arrays are often preferable for reasons such as ease of implementation. A perfect binary array with elements $s(x, y) \in \{1, -1\}$ can only exist for an area $N_x N_y$ which is a square number, because it follows from (21) that the energy of any perfect array equals to the square of its mean value.

Further, the area also has to be an even number so that the sum of the products of $+1$ and -1 in all PACF terms may disappear.

Two perfect binary arrays of size $2 \cdot 2$ and $4 \cdot 4$ are known from literature [29]. Seven more arrays of sizes $2 \cdot 8$, $6 \cdot 6$, $8 \cdot 8$, $4 \cdot 16$, $3 \cdot 12$, $6 \cdot 24$, and $12 \cdot 12$ have recently been constructed together with many more perfect binary arrays of higher dimensions [30, 31]. Such arrays of size $2 \cdot 8$, $6 \cdot 6$, and $12 \cdot 12$ are shown in Fig. 2(b).

More is known about the synthesis of binary arrays with the PACFs having a constant value of -1 between their main peaks. These arrays are called "perfect maps" or "Gordon arrays" [15, 29, 32–34]. They may be constructed by folding-up binary PN-sequences whose lengths can be written as products of two relatively prime numbers. In the same manner, by folding the binary perfect sequences described in Section II D one gets perfect amplitude-asymmetrical binary arrays. The energy efficiency of these arrays is always less than 1, but tends to approach 1 for large areas. Such arrays of area $3 \cdot 5$ and $5 \cdot 7$ are included in Fig. 2(a) and 4. Accordingly, perfect ternary arrays can be constructed by folding the suitable ternary sequences described in Section II D; an example is shown in Fig. 3.

Another way of constructing ternary arrays is by periodic multiplication of two ternary sequences or arrays, or of a ternary array with an amplitude-symmetrical binary perfect array. Likewise, the periodic product of two amplitude-asymmetrical binary arrays gives a quaternary array, etc. The number of nonzero elements in a perfect ternary array $s(x, y) \in \{1, 0, -1\}$ is a square number T; this can be seen by considering (22). Because this square number must be less than the area of the array, the energy efficiency of a ternary array is bounded by

$$\eta \leq \frac{(\text{entier } \sqrt{N_x N_y})^2}{N_x N_y} . \tag{29}$$

This bound is sketched in Fig. 4.

IV. ONE- AND TWO-DIMENSIONAL CYCLIC ORTHOGONAL FUNCTIONS

Perfect sequences and arrays find further applications in the construction of cyclic orthogonal functions. These functions are easily obtained by writing down all cyclic shifted versions of a perfect sequence or array, since all their correlation coefficients are zero by definition. For example, a perfect ternary sequence of length $N = 13$ with its cyclic shifts gives the cyclic ternary orthogonal matrix of order 13 [35]. This also applies to perfect arrays. Each perfect array of area $N_x N_y$ and its cyclic shifted versions form a system of $N_x N_y$ two-dimensional orthogonal functions [36]. These orthogonal functions are nearly binary if the perfect sequences or arrays have a large energy efficiency. Furthermore they are simple to implement due to their cyclic property. Such orthogonal functions find applications, e.g., in transform coding of pictorial data [37, 38] or in the synthesis of complementary codes [39].

V. CONCLUSION

It has been shown that there exist good constructive methods for the synthesis of perfect periodic sequences and arrays with high-energy efficiencies. This synthesis is based mainly on a search in the frequency domain and on a multiplication theorem for the periodic correlation functions of periodic sequences and arrays. These methods can be extended to higher dimensions, or to the synthesis of sets of sequences and arrays with ideal vanishing periodic crosscorrelation functions.

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